Fundamentals of Data Science

Mid 1 Imp questions

Unit I – Introduction to Data Science

1. Define Data Science. Explain its scope and importance in various fields.
2. Explain the history and evolution of Data Science.
3. Explain the major applications of Data Science.
4. Explain the Data Science Lifecycle with a neat diagram and describe each stage.
5. Compare arrays, lists, matrices, and Data Frames with suitable examples.
6. Explain the important tools and technologies used in Data Science.
7. Write a Python program to import a dataset using Pandas and perform basic exploration using head (), tail (), shape, info (), and describe ().

Unit II – Data Collection and Pre-processing

8. Explain different methods of data collection with suitable examples.
9. What is data pre-processing? Explain the major steps involved in data pre-processing.
10. Explain data cleaning techniques with examples.
11. Explain different techniques for handling missing values with an example.
12. Explain normalization and standardization. Compare them with suitable examples and formulas.
13. What is feature engineering? Explain different feature engineering techniques with examples.
14. Explain feature scaling with suitable example.
15. Explain data integration and data aggregation with suitable example.
16. For a given dataset, calculate the mean, median, mode, and standard deviation. Explain what each statistic represents.
17. Explain how to identify and handle outliers in a dataset using suitable techniques such as filtering and discretization.